Surface-Adaptive Robotic Foot with Acoustic Terrain Classification for Enhanced Locomotion

Vincent Gu, Linjie Zhang, Madhav Anugopal, Elton Lobo

***Abstract*—Legged robots operating on mixed surfaces often suffer reduced mobility due to non-adaptive foot designs. Inspired by variations in paw stiffness and morphology observed in the bear family, this work investigates whether adapting foot properties to terrain conditions can improve locomotion performance. An experimental test rig was developed, consisting of a robotic leg with a sole-switching mechanism and interchangeable terrain-specific foot pads. The system was evaluated across single-terrain and mixed-terrain conditions. Results demonstrate improved walking speed and gait efficiency when using adaptive soles compared to a single rubber sole baseline. These findings suggest that terrain-adaptive foot design can enhance mobility across diverse surfaces.**

***Index Terms*—terrain classification, acoustic sensing, legged robots, adaptive foot**

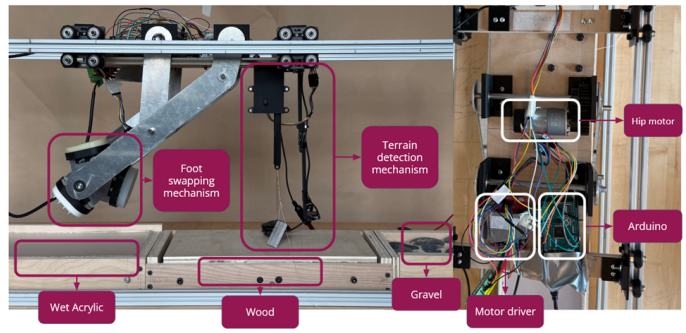

Fig. 1. Experimental test rig for evaluating single soles and terrain-adaptive sole switching.

I. Introduction

Legged robots operating on varied and unstructured terrain often suffer from reduced mobility due to insufficient traction and stability [1], [2], [3]. This limitation is critical in applications such as search and rescue, where legged robots must traverse diverse environments. A major contributor to this issue is the widespread use of non-adaptive foot designs, which are unable to adjust to different surfaces, resulting in decreased locomotion efficiency due to slipping.

Biological systems provide insight into how traction can be maintained across diverse substrates. In particular, bear species occupy a wide range of habitats, from Arctic sea ice to tropical forests. Despite sharing similar underlying paw structures, variations in papillae and wrinkle depth have been observed across species [4]. While direct comparisons of paw stiffness are limited, prior work has shown that softer backing materials can increase conformability, leading to greater contact area and friction [5].

Prior robotic approaches have explored various strategies to improve traction, including granular jamming feet [6], passive adaptive feet [7], and bio-inspired designs [8], [9], [10]. However, these designs typically focus on a single aspect of foot performance.

In this work, we investigate the combined effect of material stiffness and surface morphology on traction across different substrates. We hypothesize that terrain-adaptive foot design improves locomotion efficiency compared to non-adaptive designs.

II. Design

To test the performance of each sole and combinations of different soles, we developed a test rig consisting of a robotics leg with a foot swapping mechanism, mounted on an aluminum frame. As shown in Fig. 1, the system also includes a terrain detection module and three interchangeable test surfaces: wet acrylic, wood, and gravel.

A. Main Mechanical Structure

A Chebyshev linkage was employed in the leg design. This linkage system allows the robot to mimic a human walking gait with a single motor, which reduces control complexity compared to multi-actuator leg mechanisms. The links are fabricated out of aluminum for structural rigidity, and have lengths of 60 mm, 150 mm, and 300 mm, with the pivots of the top two links spaced 120 mm apart. This configuration produces a stride length of approximately 100 mm under no-slip conditions.

The linkages are mounted on a wooden chassis, which also supports the hip motor, the terrain detection module, and other electronic components. The wooden chassis are mounted onto the aluminum frame with four sets of extrusion rollers, which allows the leg assembly to move along the aluminum frame while walking.

B. Foot Swapping Mechanism

As demonstrated in Fig. 2, the sole switching mechanism consists of a triangular plate onto which three soles are mounted, The plate is press-fitted onto a steel shaft driven by a DC motor (Pololu 25D, 12 V, 4.4:1 gear ratio) through a worm gear set. The worm gear provides non-backdrivable transmission, allowing the selected sole to remain securely locked in place during locomotion.

C. Electronics

A dynamics model of our leg was developed using the self-manipulation framework developed by Professor Aaron

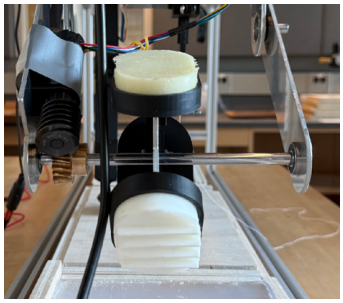

Fig. 2. Sole-switching mechanism with three mounted soles driven by a worm gear.

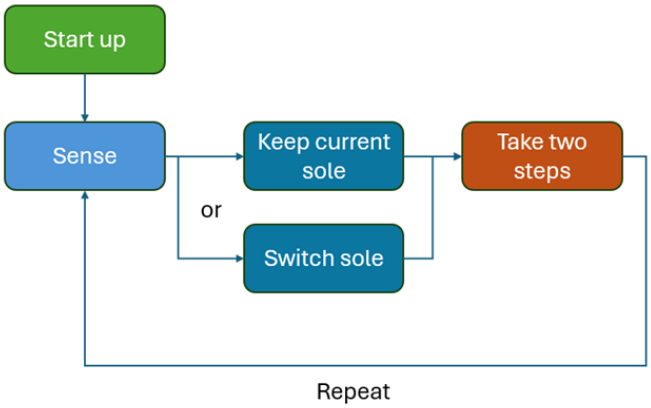

Fig. 3. Control logic for terrain-adaptive locomotion with automatic sole selection.

Johnson [11] (See Supplement Materials S1). We performed dynamic analysis of our leg when the foot is in contact with the ground as we assume that is when the motor needs to provide the highest amount of torque. We estimated the torque at the hip joint to be approximately 1.07 N·m (with a safety factor of 1.3), and about 0.002 N·m to rotate the switching plate. We selected Pololu 37D 12 V motor with 50:1 gear box with a stall torque of 2 N·m as our hip motor, which is close to twice of our anticipated torque. We also selected a Pololu 25D motor with a 4.4:1 gear box as our switch plate motor, as it is the smallest Pololu DC motor with encoder.

As shown in Fig. 1, A RoboClaw 2×15A motor controller is used, as it supports direct integration with quadrature encoders and enables closed-loop PID control of the hip and switch plate motors at the driver level, thereby reducing computational load on the micro controller. An Arduino Mega is used for control logic. The arduino communicates with the motor driver via TTL, and interfaces with an external laptop running the terrain classification algorithm through a serial connection.

A third motor (Pololu 25D) is used in the impact mechanism for the acoustic terrain classification subsystem. This motor is driven by a TB6612 motor driver, with PID control implemented on the Arduino.

The electronics system is powered by an external 12 V supply via a tethered connection, and does not include an onboard battery.

D. Acoustic Terrain Classification Subsystem

The acoustic terrain classification subsystem identifies terrain types from impact-induced acoustic signals and provides real-time input for the adaptive robotic foot. Instead of relying on passive environmental vibrations, the system actively generates controlled impact events with an impact mechanism, producing short, high energy signals that exhibit clearer material-dependent characteristics and are less sensitive to low-frequency disturbances.

A motor-driven impact mechanism is used to periodically excite the terrain surface. The design enables repeated lifting and release of an impact mass by using a rack-and-pinion mechanism. The pinion is partially toothed, such that engagement with the rack occurs only over a portion of its rotation, enabling the mass to be lifted during engagement and passively dropped once the teeth disengage. This mechanism generates consistent and repeatable contact events for both data collection and real-time terrain classification. Compared to relying on footstep-induced vibrations, this approach provides more reliable and controllable excitation, reducing variability introduced by gait dynamics.

Structure-borne vibrations are captured using a contact-based sensing module mounted on the robot chassis. The sensor maintains continuous contact with the terrain, allowing direct measurement of local material responses. The recorded signals are processed using standard acoustic feature extraction methods for classification. By focusing on localized impact responses rather than global vibration propagation, the system demonstrates improved robustness under environmental noise and structural interference.

E. Control Logic

Fig. 3 illustrates the control logic of the test rig when used with automatic terrain sensing. The system begins by sensing the terrain, once the type of terrain has been sensed the algorithm will compare the terrain to the current sole type. If the terrain matches with the current sole, the switching mechanism will not be actuated. Otherwise, the switching mechanism will be actuated to switch to the matching sole. Following this decision, the robot executes two steps before returning to the sensing stage. See the supplementary videos for demonstration.

It should be noted that the automatic sensing mechanism was disabled during all experiments for reliability, where the terrain type was manually passed to the robot via serial communication.

III. Methods

A. Terrain-Specific Sole Material Selection

An initial collection of materials was shortlisted from the Granta database with a focus on three core pillars: functional requirements, production feasibility, and economic viability. Functional requirements, such as the ideal material properties required for the three terrains (gravel, wood, and wet acrylic), narrowed the material selection list to 50 materials. It was further reduced to 25 using production feasibility as

a benchmark, and then to 10 using economic viability, which considered material sourcing, cost, and in-house assembly (see Supplemental Material S3).

The materials were then evaluated using ANSYS Workbench. A model focusing on the worst-case scenario of a robot slipping while in contact with the ground was developed and tested across the 10 candidate materials. This helped identify the top two materials for each terrain, narrowing the list to six materials (see Supplemental Material S4).

B. Terrain-Specific Sole Morphology Selection

As shown in Fig. 4, passive sole morphologies were designed separately for wood, wet acrylic, and gravel, as each terrain presents a different contact condition. Wood was treated as a dry rigid surface where maintaining contact area is critical, wet acrylic as a lubricated rigid surface requiring fluid evacuation, and gravel as a loose granular surface where penetration and interlocking are beneficial.

Candidate tread patterns were selected from biological and bio-inspired examples and evaluated using terrain-specific weighted decision matrices. The final morphology set consisted of flat, herringbone, and lamellae for wood; siped grid, deep herringbone, and hex cells for wet acrylic; and block lugs, grouser ribs, and chevron for gravel.

For physical material screening, one morphology per terrain was selected to isolate material effects: flat for wood, siped grid for wet acrylic, and grouser ribs for gravel. In each case, three materials were tested: urethane rubber, Dragon Skin, and polyurethane.

C. Experimental Protocol and Data Collection

Experiments were organized into two stages: E1 (single-terrain material screening) and E2 (adaptive versus fixed multi-terrain evaluation). E1 was used to narrow terrain-specific sole choices, while E2 evaluated whether terrain-adaptive sole switching improves locomotion relative to a fixed sole.

In E1, the robot was tested on one terrain at a time using one terrain-appropriate morphology while varying only the sole material. Wood used a flat sole, wet acrylic used a siped-grid sole, and gravel used a grouser-rib sole. Each material was tested for five trials per terrain, resulting in 45 total runs.

In E2, the adaptive system was compared against a fixed flat-sole baseline across three mixed-terrain courses: wood + wet acrylic, wood + gravel, and gravel + wet acrylic. In the fixed condition, a flat urethane rubber sole remained in contact throughout the run. In the adaptive condition, terrain-specific sole configurations from E1 were used. Five trials were conducted per condition per terrain pair, resulting in 30 total runs.

All locomotion trials were recorded using a Samsung S22 smartphone at a resolution of 1920 × 1080 pixels and 30 fps. Video data were used to extract traversal time, slip events, and cycle counts. Terrain preparation was standardized between trials, and any stall lasting 5 s or more was classified as a failure.

The primary metrics were average speed, cycles per distance, and slip count. E1 results were analyzed using Kruskal–Wallis tests with Dunn post-hoc comparisons, while E2 used two-tailed Mann–Whitney tests at a significance level of $\alpha = 0.05$.

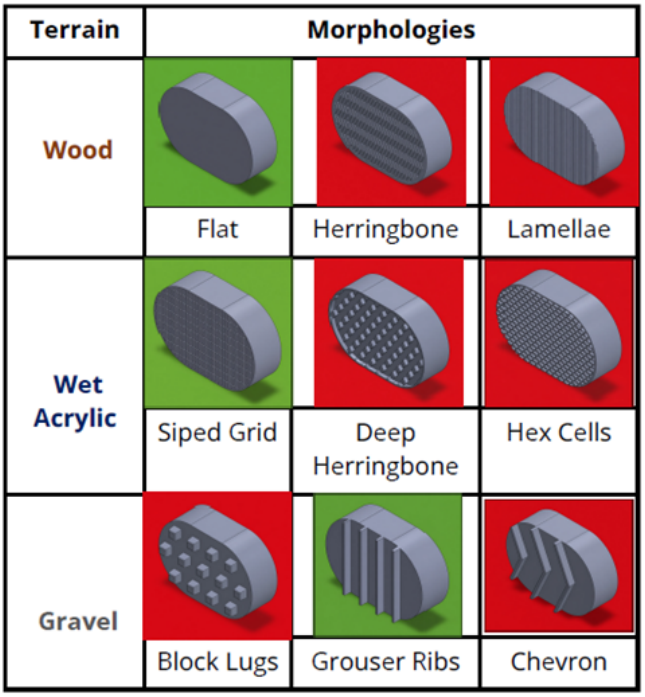

Fig. 4. Terrain-specific passive sole morphologies considered. Candidate morphologies were selected separately for wood, wet acrylic, and gravel. The ones shown in green were selected after comprehensive FEA morphology/material data analysis (see Supplemental Material S5), and used in actual experimentation

D. Acoustic Data Collection and Training Pipeline

The dataset was collected by recording impact-induced acoustic signals across different terrains, including wood, acrylic, and gravel, along with additional sliding data to capture non-impact noise observed during locomotion. Multiple recordings were taken at different spatial positions to improve robustness and reduce overfitting.

Raw audio was processed using an event-based slicing pipeline. Impact events were detected from the continuous signal using an amplitude envelope with an adaptive threshold. Key parameters, including detection thresholds, temporal windows, and minimum event spacing, were empirically tuned to balance detection sensitivity and robustness to noise.

Short segments around detected peaks were extracted as individual samples. Each sample was converted into a feature vector using acoustic descriptors such as MFCCs and spectral features. A Random Forest classifier was trained due to its robustness on small datasets and efficiency in real-time inference.

During deployment, predictions were filtered using confidence thresholds and short-term voting. To mitigate errors caused by sliding noise, classification updates were restricted to occur only between gait cycles, improving decision stability during locomotion.

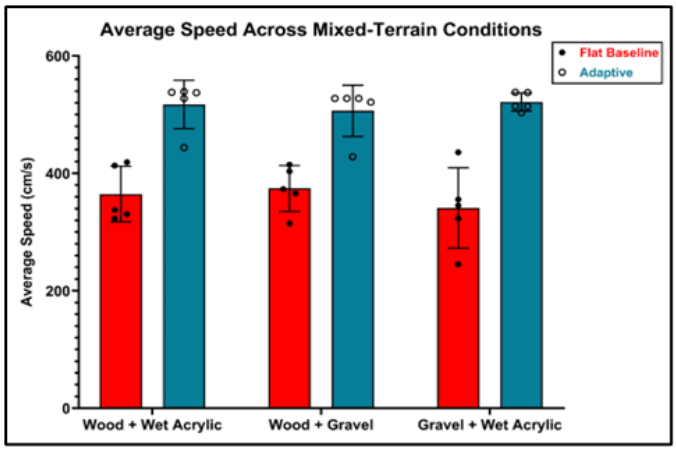

Fig. 5. Average speed across mixed-terrain conditions for fixed flat-sole baseline and adaptive sole-swapping system. Adaptive speed was significantly higher for wood + wet acrylic, wood + gravel, and gravel + wet acrylic (two-tailed Mann–Whitney, $p = 0.0079$ for all three comparisons).

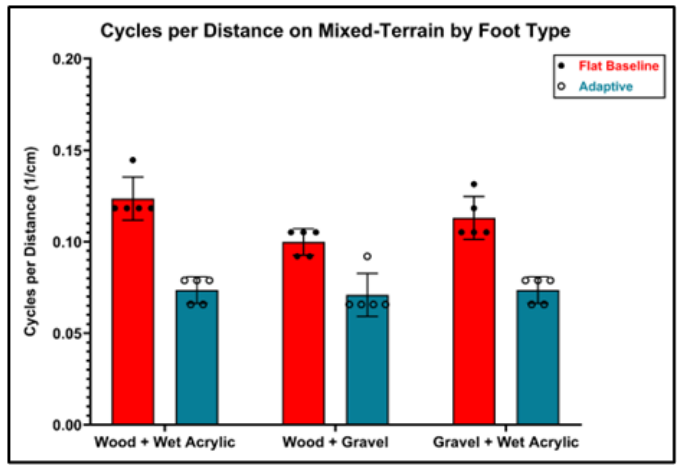

Fig. 6. Cycles per distance across mixed-terrain conditions for fixed flat-sole baseline and adaptive sole-swapping system. Lower values indicate greater gait efficiency. Adaptive cycles per distance were significantly lower for wood + wet acrylic ($p = 0.0079$), wood + gravel ($p = 0.0238$), and gravel + wet acrylic ($p = 0.0079$).

IV. Results

A. Single-Terrain Material Screening (E1)

E1 was used to screen terrain-specific sole material choices before multi-terrain testing. Rather than identifying a statistically unique winner for every terrain, it helped eliminate weak options and guide final selections using both statistical analysis and observed locomotion behavior. The selected materials were Urethane Rubber for wood, Dragon Skin for wet acrylic, and Polyurethane for gravel, with the clearest statistical separation occurring on wet acrylic (see Supplemental Material S6).

B. Adaptive Versus Fixed Multi-Terrain Evaluation (E2)

The strongest evidence in the study came from E2, where the adaptive sole-swapping system was compared against a fixed flat-sole baseline on mixed-terrain courses. This stage directly tested the project's main question: whether terrain-specific sole adaptation improves locomotion when the terrain changes.

Across all three terrain pairs, the adaptive condition produced higher average speed than the fixed baseline and was significantly faster on wood + wet acrylic, wood + gravel, and gravel + wet acrylic (two-tailed Mann–Whitney, $p = 0.0079$ for all three comparisons; Fig. 5).

A similarly consistent pattern was observed for cycles per distance. The adaptive condition required fewer gait cycles per unit distance, with significant differences on wood + wet acrylic ($p = 0.0079$), wood + gravel ($p = 0.0238$), and gravel + wet acrylic ($p = 0.0079$; Fig. 6). These lower cycles-per-distance values indicate improved gait efficiency in addition to higher speed.

Taken together, E2 provides the clearest support for the project hypothesis: under the tested conditions, the adaptive robot consistently traversed mixed-terrain courses faster and more efficiently than the fixed baseline (see Supplemental Material S6).

TABLE I
Classification performance by terrain class on the sliced dataset (pilot evaluation)

Class	Precision	Recall	Interpretation
Acrylic	0.99	0.90	Confusion with wood
Wood	0.97	0.98	Stable classification
Gravel	0.84	1.00	Absorbs sliding misclassifications
Sliding	1.00	0.61	Frequently misclassified as gravel

C. Acoustic Classification Model

The trained classifier achieved an overall test accuracy of 93.22% on the sliced dataset, indicating that impact-induced acoustic signals contain sufficient material-specific information for terrain discrimination under controlled conditions. The event-based slicing pipeline contributes to robustness by focusing on short, high-energy signal segments and reducing background noise.

Performance was sensitive to slicing parameters, which were empirically tuned to balance detection sensitivity and noise robustness. High recall was achieved for impact-based terrain classes, particularly for wood (0.98) and gravel (1.00). However, confusion was observed between acoustically similar materials such as wood and acrylic. A sliding class was introduced to account for non-impact signals observed during locomotion; however, its performance remained limited (recall = 0.61), with most errors corresponding to misclassification as gravel.

This limitation is attributed to both feature overlap and a mismatch between the impact-oriented slicing pipeline and the non-transient nature of sliding signals. To mitigate this issue at the system level, terrain classification updates were restricted to occur only between gait cycles, improving decision stability during locomotion. As the dataset is limited and collected under controlled conditions, these results should be interpreted as a pilot evaluation of feasibility.

Sliding exhibits the lowest recall and is primarily confused with gravel, indicating a limitation of the impact-based feature

representation.

V. Conclusions and Future Work

A. Conclusion

This study investigated whether terrain-adaptive foot soles can improve locomotion relative to a fixed non-adaptive foot. The strongest evidence came from the E2 multi-terrain experiments, where the adaptive sole-swapping system showed higher average speed and lower cycles per distance than the fixed flat-sole baseline across all tested terrain pairs.

E1 played a supporting role by narrowing the design space and identifying practical terrain-specific sole material selections. Overall, the main takeaway is that terrain-specific foot adaptation is a promising strategy for improving legged-robot mobility on changing terrain.

B. Future Work

Future work will focus on improving the mechanical and experimental capabilities of the platform. A suspension mechanism will be incorporated to provide vertical compliance, reducing sensitivity to terrain height variations and improving robustness during locomotion. The length of the test rig will be extended to accommodate more test surfaces simultaneously, enabling more comprehensive evaluations. In addition, new soles will be designed for more terrain types. Finally, the terrain sensing module will be systematically evaluated to quantify classification accuracy and its impact on overall system performance.

From the sensing perspective, future efforts will focus on improving data diversity and signal consistency, particularly for challenging terrains such as gravel. Integrating sensors directly into the robot soles, rather than using a dragged probe, is expected to improve alignment between sensing and contact conditions while reducing sliding-induced noise. Furthermore, the terrain sensing module will be systematically evaluated to quantify classification accuracy and its impact on overall system performance, and its deployment on embedded hardware will be explored to enable fully untethered operation.

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Data and Code Availability

The Arduino control programs and acoustic terrain classification code are available at:

https://github.com/Vincent1029384756/24775_solemate

The MATLAB code for dynamics analysis can be found at:

https://drive.mathworks.com/sharing/55fda9f1-ea20-4098-8e23-37d4c4c76b8e/

The MATLAB code for kinematics analysis can be found at:

https://drive.mathworks.com/sharing/463d7df3-4c95-40e0-88d8-e54ca89d29a1/

Experimental data on which statistical analysis was conducted:

https://docs.google.com/spreadsheets/d/1kMbG9yM2OXjmHQoffdqNX1-xY1iwyrwW/edit

https://docs.google.com/spreadsheets/d/1QOuC4dFbLt6Uzmrlkjt54H0wTFoXkAX-/edit

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Appendix A: Individual Contribution

Vincent Gu: Contributed to the dynamics analysis, mechanical design, and electronics integration of the system. Developed a dynamic model of the leg linkage using a self-manipulation framework and used the resulting torque estimates for motor selection. On the mechanical side, contributed to the structural design of the robotic leg, including linkage geometry and overall system layout. Co-designed the sole-switching mechanism with Linjie, fabricated the aluminum linkages with assistance from TechSpark staff, and produced all 3D-printed components, including mounting brackets, mold, and sole mounts. Worked on system assembly with Linjie. On the electronics side, independently designed the system architecture and selected all components. Implemented Arduino-based control for the hip and sole-switching motors, and later integrated the control system with the terrain sensing

module. Performed testing, debugging, and tuning. Operated the leg during all experiments.

Linjie Zhang: I worked on the mechanical design and acoustic sensing components of the system. On the hardware side, I contributed to the structural designs and integration of the robotic platform, including the aluminum V-slot frame, chassis layout, and the sole-switching mechanism. This involved component dimensioning (e.g., shaft length and bearing fit) and ensuring proper alignment and tolerances for reliable assembly. I also assisted in mechanical integration tasks such as press-fit assembly of shafts and bearings and supported the overall assembly of the robotic leg system.

On the sensing side, I independently developed the acoustic terrain classification pipeline. Preliminary feasibility experiments using a free-fall impact setup with a contact microphone were conducted to validate the approach prior to system integration. I then designed the impact generation mechanism and developed an impact-based data collection and event-driven signal processing framework, including an adaptive slicing method with empirically tuned parameters to balance detection sensitivity and noise robustness. I extracted acoustic features (e.g., MFCC and spectral features) and trained a Random Forest classifier achieving 93.22

I also collaborated on system-level integration by refining the control logic to incorporate classification output. This included implementing filtering strategies such as confidence thresholds, short-term voting, and gait-cycle-based gating to improve robustness during locomotion. This work enabled reliable terrain recognition and contributed directly to the adaptive sole-switching behavior of the system.

Elton Lobo: I worked on the kinematic analysis, terrain-specific sole morphology selection, experimental methodology, and statistical evaluation of locomotion performance. I developed the MATLAB kinematic model of the Chebyshev Lambda leg linkage, including the loop-closure formulation, full-cycle numerical simulation, and extraction of joint and foot-point kinematics for the final geometry. This was used to verify stride and stance behavior and to provide representative stance-phase inputs for dynamics analysis. On the morphology side, I did the full literature review and narrowed down terrain-specific tread concepts using weighted decision matrices, completed the SolidWorks models for the selected sole morphologies, and also designed the reusable mold system used for casting the sole used in experimentations. I also contributed to shared fabrication and terrain-preparation tasks. During experimentation, I structured the E1 single-terrain screening and E2 adaptive-versus-fixed testing methodology, guided the experiment flow, and assisted in data collection. After data collection, I independently performed the GraphPad Prism statistical analysis and generated the grouped performance plots that supported the project's main result: the adaptive sole-swapping system outperformed the fixed flat-sole baseline on mixed terrain.

Madhav Anugopal: I contributed to design and initial calculations of the linkage system and the sole rotation plate. I developed the initial flat sole cad file and then also completed the material selection process. I spent a long time designing and executing the worst case scenario FEA for both materials and then modified it to work for a combination of different materials and morphology.

Using the FEA results and the mould designed by my teammates I cast 9 sets of soles with 3 different materials and 3 different morphologies. This process involved sealing the mold and mixing the 2 part material combination. Pouring the solution in a steady and slow manner to avoid the formation of air bubbles and then demoulding them at a later date. Demoulding was harder than the actual pouring with it being focused on removing the sole without damaging both the sole and the mold as we were ideally planning to reuse as many molds as possible by doing each set of soles in batches of 3 for each of the 3 materials. I also then assembled the sole to the sole holder designed by another teammate. I supported the team during the first round of experiments and then spent a good amount of time in data extraction and extrapolation. The excel sheet with the raw data and data processing equations was developed by me.

From a more holistic perspective a lot of my contributions were helping the team in assembly or problem solving. I supported the team by helping with the linkage fabrication and when required the robot frame. I also supported the team by helping fabricate the boxes which contain our 3 different terrains. I supported the team by working on the supplementary documentation for the FEA analysis earlier so that it was fully compiled well ahead of when the final report was due. I also helped the team in keeping track of deliverables and our pace making sure we had results compiled before the poster was due so that we could work on having a good poster and working robot.

AI Usage

Chatgpt and Claude used for grammar and revising.